

Appendix H. Additional Failure Modes and Qualitative Examples

QID	Category	Question Type	Confidence Method	Question	Model	Support	Prediction	Correct Answer	Confidence	Interpretation
qid_2061	High support + wrong	complex	token_logprob	Under which film genre would Disturbed be classified?	OLMo32	61,482	nu metal and alternative metal	horror film	0.48	The model ignores the relation constraint ('film genre') and instead defaults to the dominant semantic association of Disturbed as a musical band.
qid_11134	High support + wrong	complex	token_logprob	To which religious tradition is Andrew the Apostle associated, specifically in relation to the Catholic Church?	OLMo32	33,580	Jew	Catholic Church	0.57	The model correctly answers the template-based version but fails under the more complex formulation.
qid_3687	Low support + correct	simple	self_consistency	What country is Kondh, Surendranagar in?	OLMo32	0	India	India	1.00	The model infers the answer despite no retrieved supporting evidence.
qid_11377	Low support + correct	simple	self_consistency	What sport is the 2014–2015 Premier League of Bosnia and Herzegovina associated with?	OLMo32	0	Football (soccer)	association football	1.00	The model appears to rely on compositional reasoning from contextual cues.
qid_636	Low support + high confidence	complex	verbalized_confidence	Could you tell me the name of the city that served as Ivan Khomukha's birthplace?	Amber	0	Moscow	Ipatovo	0.95	The model confidently predicts a prominent city despite lacking detectable supporting evidence.
qid_675	Low support + high confidence	complex	verbalized_confidence	Can you tell me the city that serves as the birthplace of Ramón Luis Valcárcel Siso?	Amber	0	Madrid	Murcia	0.95	The prediction is semantically plausible but unsupported by retrieved evidence.

QID	Category	Question Type	Confidence Method	Question	Model	Support	Prediction	Correct Answer	Confidence	Interpretation
qid_3665	High support + low confidence	complex	self_consistency	Carson River is located within the borders of which country?	OLMo	13,513	The Carson River flows through the U.S. states of Nevada and California	United States of America	0.00	The model answers correctly despite extremely low self-consistency confidence.
qid_11192	High support + low confidence	complex	self_consistency	To which religious tradition does Marian University belong?	OLMo	8,314	Roman Catholic Church	Catholic Church	0.00	The answer is semantically correct but the model remains highly uncertain.